INCH-POUND
MIL-DTL-50374B
14 September 2004
SUPERSEDING
MIL-I-50374A
11February 1993

DETAIL SPECIFICATION

INITIATOR, PROPELLANT ACTUATED, M104 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M104 Propellant Actuated Initiator Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-2550	Ammunition, General Specification For
MIL-P-46994	Charge, Propelling, 8-Inch, M1, (Green Bag) Loading, Assembling and Packing

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (Code 4230), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, or emailed to pennam@ih.navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

AMSC N/A FSC 1377

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MIL-DTL-50375 Initiator, Propellant Actuated, M104 Metal Parts Assembly

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

DL10551446	Initiator, Propellant Actuated, M104, Assembly
9252247	Carton, Packing Ammunition for, Initiator Propellant Actuated, M104
9252248	Box, Packing, Ammunition, for Initiator, Propellant Actuated, M104
9252249	Carton, Alternative, Packing Ammunition, for Initiator, Propellant Actuated, M104

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Technical Information Branch (Code 4230), Indian Head, MD 20640-5035.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E1742 Standard Practice for Radiographic Examination

(Copies of this document are available online at http://www.astm.org/cgi-bin/SoftCart.exe/STORE/standardsearch.shtml?L+mystore+afpt6301+1062008944 or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

AMERICAN NATIONAL STANDARD INSTITUTE/AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Zl.4	Sampling Procedures and Tables for Inspection by Attributes
ANSI/ASQ A8402	Quality Management and Quality Assurance - Vocabulary

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(Copies of these documents are available online at http://webstore.ansi.org/ansidocstore/default.asp, or from the American Society for Quality, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the initiators shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2 Conformance to documents. The M104 Initiator covered by this specification shall be manufactured in accordance with DL10551446.

3.3 Components.

3.3.1 Primary components. For the purpose of this specification, propellant, II-A Bi-convex pellets (MIL-P-46994) is defined as a primary component. Only primary components from a single lot shall be used in a production lot of initiators; however, one primary component production lot may be used in more than one initiator production lot.

3.3.2 Component parts. All component parts shall meet the applicable requirements of MIL-DTL-50375 prior to assembly.

3.4 Breakaway torque. The female body and male body of each assembled initiator shall withstand a breakaway torque application of 400 inch-pounds without loosening or deformation (see 4.3.1).

3.5 Residual magnetism. The initiator shall not deflect the indicator of the compass more than five degrees in either direction (see 4.3.2).

3.6 Radiographic examination. Each initiator assembly shall be radiographically examined for defects (see 4.3.3).

3.7 Leak test. Each assembled initiator shall exhibit no air leakage greater than 1×10^{-4} standard cubic centimeters per second (see 4.3.4) at a 14.7 psia pressure differential.

3.8 Functional.

3.8.1 Ballistics. Initiators loaded with accepted pyrotechnic components (II-A pellets) and temperature conditioned shall produce pressure values specified in table I. Pressure values shall be recorded in a 6.5-cubic-inch closed bomb, Drawing 838AS163.

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3.8.1.1 Initiation. The M104 initiators shall be assembled and fired with JAU-2/A25 initiator firing mechanisms complete with firing pins, shear pins, PVU-1/A ignition devices, and ignition device retainers. Minimum gas pressure is 800 psig for actuation.

3.8.2 Misfires. There shall be no misfires.

3.8.3 Mechanical failures. The initiator shall function without mechanical failure (see 6.4.2).

3.9 Workmanship. Initiators shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.9.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

TABLE I. Pressure values.

Temperature (°F)	Minimum Pressure (psig)	Ignition delay (see 6.4.1) milliseconds, max
–60 to –70	300	40
65 to 75	300	40
195 to 205	300	40

3.9.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.9.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.1.1 Quality assurance terms and definitions. Reference shall be made to ASQ A8402 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550 and ASTM E 1742.

4.2 Inspection provisions.

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4.2.1 Lot.

4.2.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation, and identification of lots shall be in accordance with ASQ Z1.4.

4.2.1.2 Size of lot. The production lot size shall be as specified in the contract (see 6.2).

4.2.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed in accordance with the classification of defects. Four power (4x) magnification may be used for inspection of such defects as burr and foreign matter. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.2.2.1 Assembly, Drawing 10551446.

Characteristics	Defect	Method of Inspection
Critical		
1	Metal defective	Visual
2	Any part missing, except as otherwise classified	Visual/X-ray
3	Improper assembly	Visual/X-ray
4	Marking missing, incorrect, or illegible	Visual
Major		
101	Thread damaged or not full	Visual
102	Burr	Visual
103	Foreign matter	Visual
Minor		
201	Protective cap missing	Visual
202	Protective finish inadequate or missing	Visual
203	Adhesive missing or inadequate	Visual

4.2.2.2 Carton, Fiberboard (Inner) Ammunition, Drawing 9252247.

Critical
None defined

Major
None defined

Minor		
201	Contents loose	Manual
202	Box damaged	Visual
203	Sealing strip broken or missing	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Cut, damaged, or improperly sealed bag	Visual

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4.2.2.3 Box, Ammunition Packing, Drawing 9252248.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual
202	Strapping missing, loose, broken, or improperly engaged	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.2.2.4 Carton, Alternative, Ammunition Packing, Drawing 9252249.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual/Visual
202	Carton broken or damaged	Visual
203	Number of units contained incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.2.3 Inspection testing. The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque. Each assembly shall be subjected to the breakaway torque application. Failure of any assembly to comply with 3.4 shall be cause for rejection of the assembly.

4.2.3.2 Residual magnetism. Each initiator shall be subjected to the test for excessive residual magnetism as specified in 4.3.2. Failure of any initiator to comply with 3.5 shall be cause for rejection of the initiator.

4.2.3.3 Functional tests. The following tests shall be performed by the Government.

4.2.3.3.1 Ballistics. Twenty-one units shall be tested to determine compliance with 3.8. The units shall be temperature conditioned as provided in table II and 4.3.5.2. Failure of any unit to comply with 3.8 shall be cause for rejection of the lot.

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TABLE II. Conditioning temperature.

Temperature (°F)	Sample Size
–60 to –70	7
65 to 75	7
195 to 205	7

4.2.3.3.2 Retest. There shall be no retests.

4.2.3.3.3 Test failure. If a test failure is attributable to an assignable cause, excluding the test initiator, the original test results shall be discarded and that test reconducted.

4.3 Test methods and procedures.

4.3.1 Breakaway torque. A torque wrench set as specified in 3.4 shall be used in conjunction with a fixture to determine the adequacy of the joint.

4.3.2 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.3.3 Radiographic examination. All assembled initiators shall be radiographically examined in accordance with ASTM E 1742, with the exception that an initiator assembly will be used for the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection. The initiators shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the X-ray machine. All initiators shall be identified with serial numbers prior to X-ray examination. The serial numbers shall be in consecutive numerical order beginning with the number 001 in each production lot. Any missing numbers are to be identified on the X-rays. The initiators shall be arranged on trays or boards in consecutive order, and each radiograph shall carry a permanent identification of the units displayed thereon. The radiographic plate identification shall include item nomenclature, the complete lot number, the span of serial numbers displayed, and the contract number. The radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. Defective initiators found during X-ray review are to be marked on the X-ray prior to shipment and those defective items shall be removed from the production lot. Film interpretation of the radiographs shall verify the presence and correct positioning of the following parts:

a. Flash tube, P/N 10551453
b. II-A Potassium nitrate pellets, MIL-P-46994
c. Sealing rings, P/N 10551451

and shall also verify that:

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d. all .062 holes are present and intersect with .116 diameter holes in flash tube, P/N 10551453
e. there is metal between the two longitudinal .116 diameter holes in flash tube, P/N 10551453

4.3.4 Leak test. Each initiator shall be subjected to specified pressure differential. The method for determining airtightness shall be with a helium mass spectrometer. The helium leak rate shall be converted to standard cubic inches air. Failure to comply with the requirements of 3.7 shall be cause for rejection of the cartridge.

4.3.5 Functional.

4.3.5.1 Ballistics. The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample unit shall be reconditioned in accordance with 4.3.5.2.

4.3.5.2 Temperature conditioning. Each assembled initiator shall be conditioned for not less than 4 hours nor greater than 24 hours. When reconditioning initiators at -60°F, all condensation shall be removed from the exterior surfaces before returning the initiator to the conditioning chamber.

4.4 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with section 5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The initiator assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems. Since the initiator assembly was developed solely for use in military aircraft, there is no commercial application.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and revision letter of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced

c. Production test activity and lot size (see 4.1 and 4.2.1)

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d. Directions for shipping radiographs of entire production lot along with ballistic sample to the activity conducting production lot acceptance tests (see 4.3.3)

e. Level of packing and marking (see 5.1)

f. That the safety precaution requirements of the ``Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of ``Ammunition and Explosives Ashore," OP 5 are applicable.

6.3 Hazard notice. The initiator described herein and certain of its components are flammable, toxic, or explosive and present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these properties and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.4 Definitions.

6.4.1 Ignition delay. Ignition delay time is defined as the time that elapses between the entrance of hot gas into the intake port of the M104 Initiator and the beginning of the continuous pressure rise on the pressure time record.

6.4.2 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.5 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.6 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes. This revision was authorized by ECP number 045220I017.

Custodian:
Navy - OS
Air Force - 99

Preparing activity
Navy - OS
(Project 1377-0223)

Review activity:
Air Force - 70

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